Final Project

Important Notes

1. Students must upload a project report named **Project_[student_id].pdf** along with the required Python code in either **.py** or **.ipynb** formats, combined into an archive file named **Project_[student_id].zip** in the Courses platform.

2. Please **select only one** of the two papers provided in this file and complete the requested tasks.

3. If cheating is discovered or suspected among students, points will be **deducted**.

4. If there is any question, you can contact us through the following email addresses:

fatemehmousavi@aut.ac.ir , es.khosravi@aut.ac.ir, aidin.khalili@aut.ac.ir, Payamzohari@aut.ac.ir, dorsa.macky@aut.ac.ir

Wish you best of luck!

Paper 1, An improved context-aware weighted matrix factorization algorithm for point of interest recommendation in LBSN

Theoretical Questions (Payam Zohari)

Part 1: Problem Definition and Core Contributions

1. The paper identifies the sparsity of user check-in data as a fundamental challenge in POI recommendation within location-based social networks. Analyze why this sparsity problem is particularly severe in LBSNs compared to traditional recommendation domains such as movie or product recommendations, and explain how the ICWMF algorithm's multi-context approach theoretically addresses this challenge.
2. The paper leverages three distinct types of contextual information. For each type, explain the underlying assumption about user behavior that justifies its inclusion, and analyze the specific role each contextual component plays in addressing the limitations of standard matrix factorization approaches.

Part 2: Context Modeling Components

1. The paper employs the Ebbinghaus forgetting curve to model temporal influence. Explain the mathematical foundation of this curve and analyze how it captures the temporal decay of user interest. What are the theoretical implications of using this model compared to simpler time-based decay functions?
2. The Gaussian model is employed to construct proximity location relationships, which is then integrated as a regularization term. Analyze the mathematical intuition behind using a Gaussian distribution for this purpose. Why does modeling proximity as a probability distribution provide a theoretically sound foundation for capturing geographical influence, and what problem does its use as a regularization term specifically address in the matrix factorization framework?
3. The paper constructs an implicit feedback term by combining geographical influence from the user perspective with social relationship modeling. Explain the theoretical justification for this construction. Why is modeling implicit feedback crucial for inferring user preferences for unvisited POIs, and how does this approach differ from relying solely on explicit feedback signals?

Part 3: Objective Function and Optimization

1. The objective function of weighted matrix factorization is reconstructed by adding an implicit feedback term and a regularization term. Analyze the theoretical rationale behind each addition. Why is the standard weighted matrix factorization objective function insufficient for POI recommendation, and how do these

modifications transform the learning process to better capture user preferences in sparse data environments?

2. Weighted matrix factorization learns two latent feature matrices. Explain the role of each feature matrix, the information they encode, and how the reconstructed objective function guides their learning. What theoretical guarantees or properties does the Alternating Least Squares optimization method offer for this type of problem, and why is it preferred over alternative optimization approaches?

Part 4: Evaluation and Theoretical Analysis

1. The paper evaluates ICWMF using precision and recall metrics. Analyze the theoretical strengths and limitations of these metrics for evaluating POI recommendation systems. Under what conditions might precision-recall be insufficient or misleading, and what alternative or complementary metrics might provide a more comprehensive evaluation framework?
2. The experimental results indicate that ICWMF outperforms four comparison methods. Beyond the specific contextual components, identify at least one significant theoretical issue or limitation with the ICWMF approach that could affect its generalizability across different LBSN datasets or user populations. Propose a theoretically-grounded improvement that could address this limitation.
3. The paper employs a multi-context fusion strategy where temporal, geographical, and social information are integrated into a single objective function. Analyze the potential challenges of this integration approach. How might the relative importance of these contextual factors vary across different recommendation scenarios, and what theoretical justification supports the current integration method as a reasonable solution?

Implementation Questions (Fatemeh Mousavi)

Implement the proposed Improved Context-Aware Weighted Matrix Factorization (ICWMF) algorithm described in the paper. Where the paper does not specify implementation details or parameter choices, make reasonable design decisions, clearly state your assumptions, and justify your choices.

Submission requirements

- Submit your implementation as a Jupyter Notebook (.ipynb).
- Organize the notebook into the four parts below.
- Include appropriate comments throughout your implementation and complete explanations in your report.

Note: You may use standard Python libraries for helper functions, but the ICWMF algorithm must be implemented from scratch. Do not use existing matrix factorization or recommendation libraries.

Part 1: Data Preparation

1. Download the Gowalla dataset (SNAP).
2. Preprocess the dataset as described in the paper:
   - Remove users with fewer than 10 check-ins.
   - Remove POIs with fewer than 10 check-ins.
3. If computational resources are limited, a sampled version of the dataset may be used. Clearly state your sampling approach.
4. Construct:
   - the user–POI frequency matrix R
   - the binary preference matrix C
5. Split the data into 80% training and 20% testing, following the procedure described in the paper.

Part 2: Context Modeling

Implement the contextual components proposed in the paper. Briefly explain how each component contributes to learning user preferences.

1. **Temporal Information**
   - Update the frequency matrix using the Ebbinghaus forgetting curve described in the paper.
   - Use the Gowalla time threshold H = 100.
2. **Geographical Information**
   - Apply DBSCAN to identify each user's activity regions. (You can use sklearn.cluster.DBSCAN)
   - Compute activity centers for each cluster.
   - Model geographical influence from the user perspective following the approach described in the paper.
   - Estimate the parameters of the power-law model using the method described in the paper.
3. **Social Relationship Modeling**
   - Build the social graph from the friendship data.
   - Compute social preference scores following the method presented in the paper.

Part 3: ICWMF Implementation

Implement the ICWMF algorithm from scratch using NumPy, SciPy, and scikit-learn. Do **not** use an existing matrix factorization library.

Your implementation should include:

- construction of the objective function proposed in the paper,
- confidence weighting,
- implicit feedback modeling,
- location regularization,
- ALS optimization using the update equations provided in the paper.

You may use the same hyperparameter values as the paper for the Gowalla dataset or justify any alternative values you choose.

Run ALS until convergence or for up to 30 iterations.

Part 4: Evaluation and Analysis

1. Generate ranked recommendation lists for each user.
2. Compute:
   - Precision@5, Precision@10, Precision@15, Precision@20
   - Recall@5, Recall@10, Recall@15, Recall@20
3. Compare your results with those reported in Figure 3 of the paper.
4. Discuss:
   - similarities and differences between your results and those reported in the paper,
   - possible reasons for any discrepancies.
   - Describe the main challenges you encountered while implementing ICWMF. Explain how you addressed these challenges and justify any implementation decisions that were not explicitly specified in the paper.
   - Identify at least one issue with the proposed approach and suggest an idea for improvement.

Paper 2, Deep matrix factorization via feature subspace transfer for recommendation system

Theoretical Questions (Aidin Khalili)

Question 1

What is the central problem addressed by this paper, and what two specific drawbacks of existing deep matrix factorization approaches does it aim to overcome? Explain why these drawbacks matter for recommendation performance.

Question 2

What network architecture is used to initialize the latent features of the items, and what structural property distinguishes it from a standard autoencoder? Also, explain the mechanism FSTDMF uses to transfer auxiliary features into the target task. Include the mathematical intuition behind the penalty term.

Question 3

Discuss the role of the confidence parameter η in FSTDMF. What does the model reduce to when $\eta = 0$, and why did experiments find $\eta = 1$ optimal?

Question 4

Identify the evaluation metrics used and summarize the headline quantitative improvement FSTDMF achieved over standard DMF, including the condition under which the maximum gain occurred.

Question 5

The paper conducts an ablation study separating the two core components. What are the two components (with abbreviations), and what did the ablation reveal about their relative contributions on small versus large datasets?

Implementation Questions (Esmaeil Khosravi)

You are expected to implement and evaluate the FSTDMF recommendation model proposed in the paper. The project focuses only on the item-attribute-information setting of the model.

1. Implementation

Implement the FSTDMF model and the DMF baseline, and apply them on the following datasets:

- MovieLens-100K
- MovieLens-1M

You are only required to reproduce the results corresponding to DMF and FSTDMF reported in the paper. Implementation of other baseline methods **is not required.** You should reproduce the corresponding results of Table 3 (MovieLens-100K) and Table 4 (MovieLens-1M) of the paper, restricted to the DMF and FSTDMF rows, and compare your obtained performance with the reported values.

2. Sensitivity Analysis

a. Latent Feature Dimension (r)

- On the MovieLens-1M dataset only, conduct parameter sensitivity experiments for both the FSTDMF and DMF models, varying the latent feature dimension (r). (Testing at 3 values of r is sufficient.)
- Reproduce the analysis corresponding to Figure 6(a), Figure 6(b), and Figure 6(c).

b. Number-of-Layers

- On the MovieLens-100K dataset only, conduct a sensitivity analysis on the number of hidden layers, for the proposed FSTDMF model only. (Corresponding to sections of Table 8)

In each part, compare the results with the figures of the paper.

3. Ablation Study

Conduct an ablation study on the MovieLens-100K dataset only by implementing and comparing the following four models:

- DMF
- DMF (+SA)
- DMF (+FST)
- FSTDMF

Compare the obtained results with the corresponding discussion and findings reported in the paper (corresponding rows of Table 9).

4. Result Analysis

For every experiment conducted in this project, provide sufficient analysis and justification for the obtained results, and reference the corresponding sections/equations/figures of the paper whenever appropriate.

Notes:

1. Only the item-attribute-information version of FSTDMF is required.
2. The cross-domain recommendation tasks (Task 1 and Task 2 in the paper) **are not required**, due to their additional complexity and computational cost.
3. You do not need to exhaustively evaluate all hyperparameter combinations reported in Table 2 of the paper. The following settings are suggested and sufficient:
    - Learning rate = 0.0123
    - Regularization coefficient ∈ {0.10, 0.15}
    - For all other hyperparameters, testing only the first two or three values of the sets reported in Table 2 is sufficient.